

The table of trigonometric values

Reading a ratio from an angle, and an angle from a ratio — on paper and on a calculator.

LESSON 27–28

2 × 40 MIN

GEOMETRY 8, TEMA 24

STAGE 9 · BEYOND

LESSON CLOCK

5'

12'

8'

13'

2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Read a value from a four-figure table or a calculator.
- Find the angle when the ratio is given (the inverse operation).
- Know how each ratio behaves as the angle grows from 0° to 90°.
- Estimate before calculating, and check the answer is sensible.

TEXTBOOK

National	Tema 24, pp. 56–57
Cambridge	Extension beyond Stage 9
Workbook	—

01

Explanation

Reading the table

A four-figure table gives $\sin$, $\cos$, $\tan$ and $\cot$ for every angle from 0° to 90° , usually in steps of $6'$ (six minutes of arc, one tenth of a degree). Find the row for the whole degrees and the column for the minutes.

On a calculator, make sure it is in **degree mode** (the display shows DEG). Then $\sin 37^\circ = 0.6018\dots$, $\cos 37^\circ = 0.7986\dots$, $\tan 37^\circ = 0.7536\dots$

CHECK THE MODE FIRST, EVERY TIME

In radian mode $\sin 37$ gives -0.6435 — a negative number, which is impossible for an acute angle. A negative sine is always a mode error.

The inverse direction

Given the ratio, find the angle. On the calculator this is the second function of the same key, written $\sin^{-1}$ or $\arcsin$:

$\sin \alpha = 0.5 \implies \alpha = 30^\circ \cdot \tan \alpha = 1 \implies \alpha = 45^\circ$

In a table you read backwards: find the value in the body and go out to the edge for the angle. If the exact value is not there, take the nearest — the answer is approximate anyway.

NOT A POWER

$\sin^{-1}x$ means “the angle whose sine is x ”. It is **not** $\frac{1}{\sin x}$.

How the ratios behave

A	0 °	30 °	45 °	60 °	90 °	BEHAVIOUR
sin α	0	0.500	0.707	0.866	1	increases
cos α	1	0.866	0.707	0.500	0	decreases
tan α	0	0.577	1	1.732	—	increases without limit
cot α	—	1.732	1	0.577	0	decreases

Three facts to use as checks:

- $\sin$ and $\cos$ are always between 0 and 1;
- $\tan$ passes through 1 at 45° — below that it is less than 1, above it more;
- as α grows, $\sin$ and $\tan$ grow while $\cos$ and $\cot$ shrink.

02 Worked examples

EXAMPLE 1 Find $\sin 52^\circ$ and use it to find the leg opposite 52° in a triangle with hypotenuse 20.

① $\sin 52^\circ \approx 0.788$

From the table or calculator.

② $a = c \cdot \sin \alpha = 20 \cdot 0.788$

③ $a \approx 15.8$

Check: it is less than 20 ✓ and more than half of 20, as $52^\circ > 30^\circ$ ✓

ANSWER

≈ 15.8

EXAMPLE 2 Find the angle α if $\cos \alpha = 0.4226$.

① $\cos \alpha = 0.4226$

Given.

② Read the table backwards, or press $\cos^{-1}$.

③ $\alpha \approx 65^\circ$

Sensible: the cosine is under 0.5, so the angle is above 60° .

ANSWER

$\alpha \approx 65^\circ$

EXAMPLE 3 *Without a table, decide whether $\sin 40^\circ$ is bigger or smaller than $\cos 40^\circ$.*

① $40^\circ < 45^\circ$

② Below 45° the sine is the smaller of the two.
They are equal exactly at 45° .

③ $\sin 40^\circ < \cos 40^\circ$
Check: $0.643 < 0.766$ ✓

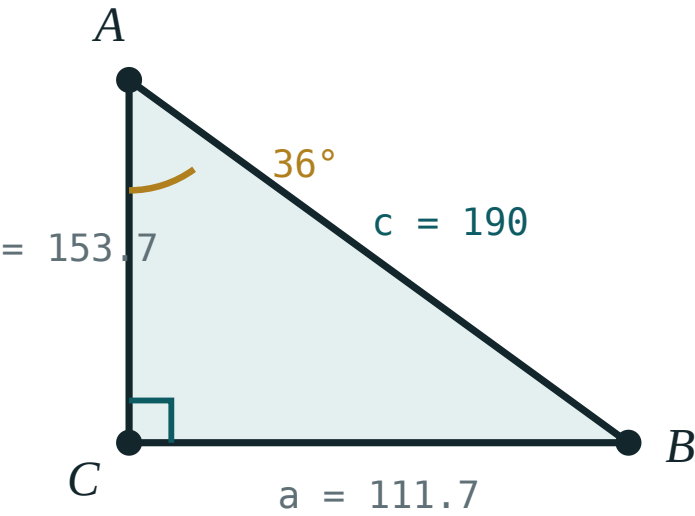
ANSWER

$$\sin 40^\circ < \cos 40^\circ$$

03 Interactive model

Sweep the angle from 10° to 80° and read the three ratios changing — sine up, cosine down.

How the ratios move with the angle



α 36°	β 54°	a (opposite) 111.7	b (adjacent) 153.7	c (hyp) 190	$\sin \alpha$ 0.59
$\cos \alpha$ 0.81	$\tan \alpha$ 0.73	$a^2 + b^2$ 36100	c^2 36100		

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Table of values	Qiymatlar jadvali	Таблица значений
Calculator	Kalkulyator	Калькулятор
Degree	Gradus	Градус
Minute (of arc)	Daqiqa	Минута
Inverse function	Teskari funksiya	Обратная функция
Increasing	O'suvchi	Возрастающая
Decreasing	Kamayuvchi	Убывающая
Round	Yaxlitlash	Округлить
Approximate value	Taqribiy qiymat	Приближённое значение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

As α grows from 0° to 90° , $\cos \alpha$:

increases

decreases

stays the same

increases then decreases

$\tan 50^\circ$ is:

less than 1

equal to 1

greater than 1

negative

$\sin^{-1}(0.5)$ equals:

2

30°

60°

0.5

A calculator gives $\sin 37 = -0.6435$. This means:

the angle is obtuse

the calculator is in radian mode

the sine can be negative

the answer is right

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Find $\sin 30^\circ$ from the table.

ANSWER 0.5

2. Find $\cos 60^\circ$.

ANSWER 0.5

3. Find $\tan 45^\circ$.

ANSWER 1

4. Find $\sin 50^\circ$ to 3 d.p.

ANSWER 0.766

5. Find $\cos 25^\circ$ to 3 d.p.

ANSWER 0.906

6. Find α if $\sin \alpha = 0.5$.

ANSWER 30°

7. Find α if $\tan \alpha = 1$.

ANSWER 45°

Medium · the standard question

7 PROBLEMS

1. Find $\sin 52^\circ$ to 3 d.p.

ANSWER 0.788

2. Find $\tan 68^\circ$ to 3 d.p.

ANSWER 2.475

3. Find α if $\cos \alpha = 0.4226$.

ANSWER 65°

4. Find α if $\tan \alpha = 0.5774$.

ANSWER 30°

5. A triangle has hypotenuse 20 and an angle of 52° . Find the opposite leg.

ANSWER ≈ 15.8

6. A triangle has hypotenuse 15 and an angle of 40° . Find the adjacent leg.

ANSWER ≈ 11.5

7. Which is bigger, $\sin 40^\circ$ or $\cos 40^\circ$?

ANSWER $\cos 40^\circ$

Hard · stretch and reason

7 PROBLEMS

1. Arrange in increasing order: $\sin 20^\circ$, $\sin 70^\circ$, $\cos 20^\circ$, $\cos 70^\circ$.

ANSWER $\cos 70^\circ = \sin 20^\circ < \sin 70^\circ = \cos 20^\circ$, so $\sin 20^\circ = \cos 70^\circ < \sin 70^\circ = \cos 20^\circ$

2. For which acute angles is $\tan \alpha > 1$?

ANSWER $45^\circ < \alpha < 90^\circ$

3. A calculator shows $\sin x = 1.2$. What has gone wrong?

ANSWER Impossible — the sine of an acute angle is under 1. A side has been divided the wrong way round.

4. A ramp has length 20 m and rises 7 m. Find its angle of slope.

ANSWER $\sin \alpha = 0.35$, so $\alpha \approx 20.5^\circ$

5. A tower is 45 m tall and its shadow is 30 m. Find the angle of elevation of the sun.

ANSWER $\tan \alpha = 1.5$, so $\alpha \approx 56.3^\circ$

6. Explain why $\tan \alpha$ has no value at 90° .

ANSWER The adjacent side would be 0, and division by 0 has no meaning.

7. Estimate $\sin 33^\circ$ without a table, using $\sin 30^\circ$ and $\sin 45^\circ$.

ANSWER Between 0.5 and 0.707, nearer the first — about 0.54. (True value 0.545.)

07 Homework

Homework — 5 problems

Geometry 8, Tema 24, pp. 56–57. Bring the table or a calculator in degree mode.

- Find $\sin 28^\circ$, $\cos 28^\circ$ and $\tan 28^\circ$ to 3 decimal places.
- Find α if $\sin \alpha = 0.8660$.
- Find α if $\cos \alpha = 0.7071$.
- A triangle has hypotenuse 25 and an angle of 36° . Find both legs.

5. Without a table, say which is bigger: $\tan 30^\circ$ or $\cot 30^\circ$. Explain.